

Comparing LDA and BERTopic for Topic Discovery in a Chinese News Corpus

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Abstract

This study compares Latent Dirichlet Allocation (LDA) and BERTopic for discovering thematic structures in a Chinese news corpus containing 5,764 documents. In addition to examining topic keywords and document distributions, the study constructs topic co-occurrence networks using Pointwise Mutual Information (PMI) and applies community detection to identify higher-level thematic structures. LDA produces 10 topics, while BERTopic produces 9 topics after topic reduction. The results show that LDA provides a more concentrated topic assignment structure, whereas BERTopic allows documents to be associated with multiple topics. The resulting topic networks also differ: the BERTopic network contains more PMI-based connections but fewer detected communities. Overall, the findings suggest that the two approaches provide complementary views of the thematic organization of the corpus.

1 Introduction

Topic modeling is a common approach for discovering latent thematic structures in large text collections. This study compares two topic modeling approaches, Latent Dirichlet Allocation (LDA) and BERTopic, on a Chinese news corpus. LDA represents documents using probabilistic distributions over topics, while BERTopic uses contextual document embeddings to identify semantically related topics.

In addition to comparing the topics identified by the two models, this study examines topic co-occurrence using Pointwise Mutual Information (PMI) and applies community detection to identify higher-level thematic structures. This provides a network-based perspective on how topics are related within the corpus.

The study addresses the following research questions:

1. How do LDA and BERTopic differ in the topic structures they identify?
2. How do the two models differ in topic co-occurrence and community structure?
3. To what extent do the topic assignments produced by LDA and BERTopic correspond to each other?

2 Dataset

The dataset consists of 5,764 Chinese-language news documents. The corpus contains articles covering a range of topics, including politics, international affairs, technology, finance, health, lifestyle, and social issues.

The documents were preprocessed before topic modeling. Text cleaning and tokenization were applied, including the removal of stopwords, punctuation, and irrelevant tokens. The

resulting tokenized documents were used as input to both LDA and BERTopic to ensure that the two models operated on the same underlying corpus.

The dataset also contains manually assigned news tags. These tags were not used as supervision during topic modeling. Instead, they were used to interpret the resulting topics and examine the relationship between discovered topics and the original categories.

3 Methodology

3.1 Latent Dirichlet Allocation

LDA was used as a probabilistic baseline for topic modeling. The model was trained with 10 topics. Each document was represented as a probability distribution over the 10 topics, and the topic with the highest probability was selected as the document’s dominant topic for subsequent analysis.

For each topic, the top words with the highest topic-word probabilities were extracted to provide an interpretable representation of the topic.

The final LDA topics were also compared with the original news tags. Topic purity was calculated as the proportion of documents within a topic belonging to its most frequent original tag.

3.2 BERTopic

BERTopic was used as an embedding-based alternative to LDA. Contextual document embeddings were generated and subsequently reduced in dimensionality before density-based clustering.

The initial BERTopic model produced 75 topics. Topic reduction was then applied, resulting in 10 topics, followed by a final topic structure containing 9 non-empty topics. The final BERTopic representation was based on the most representative words associated with each topic.

Unlike the dominant-topic representation used for LDA, BERTopic allows a document to be associated with multiple topics. Therefore, the average number of topics assigned to each document was also examined.

3.3 Topic Co-occurrence Networks

To investigate relationships between topics, topic co-occurrence networks were constructed for both models. Each topic was represented as a node. Two topics were connected when they occurred in the same document.

The strength of the association between two topics was measured using Pointwise Mutual Information (PMI):

$$PMI(x, y) = \log \frac{P(x, y)}{P(x)P(y)} \quad (1)$$

Positive PMI values indicate that two topics co-occur more frequently than would be expected under statistical independence.

Following the analysis procedure, only edges with

$$PMI > 0.05$$

were retained for the final network visualizations.

3.4 Community Detection

Community detection was applied to the PMI networks to identify groups of topics that were more strongly connected to one another.

The greedy modularity community detection algorithm was used. Each resulting community was interpreted according to the keywords of its constituent topics, allowing higher-level thematic structures to be identified beyond individual topics.

3.5 Cross-model Comparison

The topic assignments produced by LDA and BERTopic were compared at the document level. Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) were used to measure the degree of correspondence between the two topic assignments.

These metrics provide complementary measures of agreement. ARI evaluates the similarity between two clustering assignments while correcting for chance, whereas NMI measures the shared information between the two assignments.

4 Results

4.1 LDA Topic Structure

The LDA model produced 10 topics. Table 1 summarizes the main keywords, document counts, topic shares, purity values, and dominant original tags.

Table 1: Final LDA topic structure.

Topic	Top Keywords	Documents	Share (%)	Purity (%)	Main Tag
T1	網友, 分享, 好吃, 使用, 許多, 編輯	453	7.86	97.35	風生活
T2	健康, 食物, 醫師, 身體, 飲食, 維生素	467	8.10	93.15	風生活
T3	民進黨, 國民黨, 民眾黨, 立委, 總統, 選舉	1078	18.70	76.81	政治
T4	發生, 發現, 警方, 事件, 人員, 調查	666	11.55	80.48	風生活
T5	中國, 台灣, 美國, 經濟, 國家, 政府	681	11.81	48.46	國際
T6	日本, 台灣, 大學, 美國, 歷史, 旅遊	156	2.71	47.44	風生活
T7	網友, 什麼, 工作, 許多, 台灣, 這樣	960	16.66	75.94	風生活
T8	政府, 台灣, 相關, 安全, 陳建仁, 社會	349	6.05	50.43	政治
T9	工作, 申請, 獎金, 民眾, 春節, 金額	333	5.78	86.19	風生活
T10	公司, 市場, 投資, 美元, 全球, 企業	621	10.77	56.36	風生活

The largest LDA topic is T3, which accounts for 18.70% of the corpus and contains keywords strongly associated with Taiwanese electoral politics. Other relatively large topics include T7 (16.66%) and T5 (11.81%).

Topic purity varies substantially across topics. T1 and T2 have particularly high purity values of 97.35% and 93.15%, respectively, indicating strong alignment with their dominant

original tag. In contrast, T5, T6, T8, and T10 have lower purity values, suggesting greater thematic overlap with other categories.

The average topic purity across the ten topics is 69.03%.

4.2 BERTopic Topic Structure

BERTopic produced 9 final non-empty topics. Table 2 summarizes their representative keywords.

Table 2: Final BERTopic topic structure.

Topic	Representative Keywords
B0	台灣, 民進黨, 國民黨, 民眾黨, 立委, 總統, 中國, 選舉, 柯文哲, 賴清德
B1	網友, 分享, 好吃, 許多, 編輯, 討論, 責任, 推薦, 使用, 不少
B2	中國, 經濟, 市場, 美國, 美元, 公司, 投資, 日本, 全球, 台積電
B3	健康, 食物, 醫師, 飲食, 維生素, 血壓, 身體, 攝取, 營養師, 膽固醇
B4	公司, 蘋果, 影片, 使用, 用戶, 手機, 功能, 網路, 員工, 詐騙
B5	俄羅斯, 烏克蘭, 美國, 以色列, 戰爭, 德國, 普京, 國家, 納瓦爾尼, 哈瑪斯
B6	川普, 美國, 拜登, 共和黨, 初選, 總統, 北約, 海莉, 移民, 歐洲
B7	星座, 財運, 機會, 事業, 財富, 生肖, 關係, 工作, 得到, 容易
B8	投資, 錢包, 生活, 有錢人, 工作, 財富, 朋友, 人生, 理財, 金錢

The BERTopic results reveal several semantically coherent themes. B0 is strongly associated with Taiwanese domestic politics, while B5 and B6 focus on international political events involving Russia, Ukraine, Israel, the United States, and Donald Trump. B3 represents health and nutrition, whereas B4 captures technology-related content.

BERTopic also identifies more specific lifestyle and personal-finance themes, such as astrology and fortune-related content in B7 and personal finance and wealth in B8.

4.3 Comparison of Overall Model Structure

Table 3 summarizes the main differences between the two models.

Table 3: Overall comparison of LDA and BERTopic.

Metric	LDA	BERTopic
Number of topics	10	9
Number of documents	5764	5764
Average topics per document	1.00	1.96
PMI > 0.05 edges	7	15
Number of communities	3	2
Largest community (topics)	5	5
Largest topic share (%)	18.70	27.43
Average topic purity (%)	69.03	N/A

A major difference between the models is the number of topics associated with each document. The LDA analysis uses the dominant topic for each document, resulting in exactly one topic assignment per document. In contrast, BERTopic assigns an average of 1.96 topics per document under the selected topic-assignment threshold.

The largest BERTopic topic accounts for 27.43% of the corpus, compared with 18.70% for the largest LDA topic. This indicates that the BERTopic solution contains a more dominant central topic.

The topic co-occurrence networks also differ substantially. After applying the same PMI threshold, the BERTopic network contains 15 edges, compared with 7 edges for LDA. This suggests that the BERTopic topic assignments reveal more frequent or stronger topic associations under the selected network construction procedure.

4.4 Topic Co-occurrence Networks

The topic co-occurrence networks provide a complementary view of the topic structures.

For LDA, the network contains 10 nodes and 7 edges after applying the $PMI > 0.05$ threshold. Community detection identifies three communities.

For BERTopic, the network contains 9 nodes and 15 edges. The larger number of edges indicates a denser topic association structure compared with the LDA network.

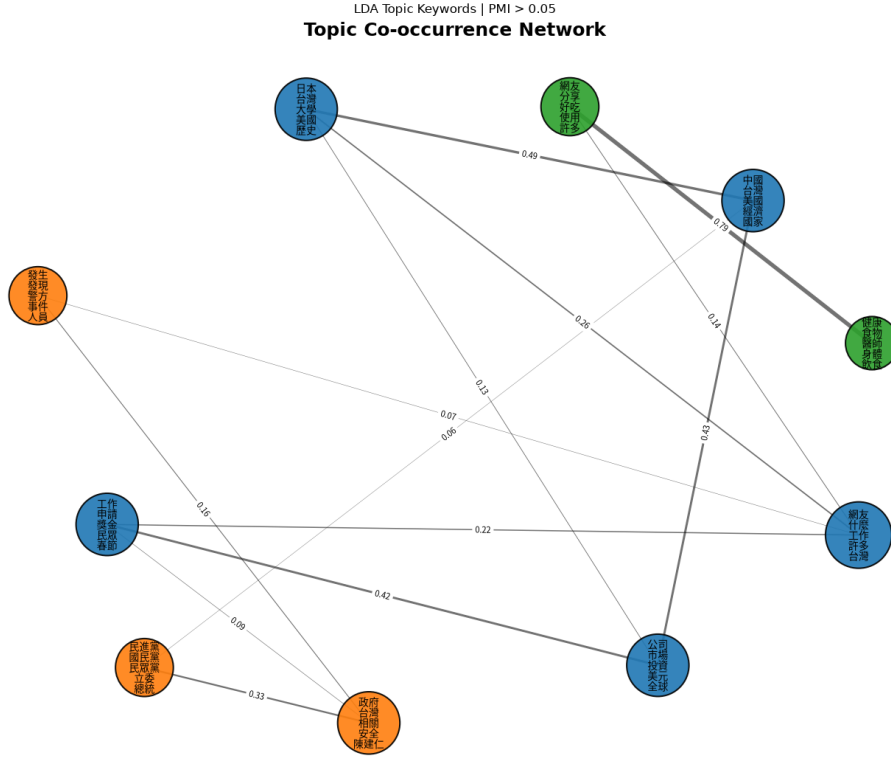


Figure 1: LDA topic co-occurrence network with $PMI > 0.05$.

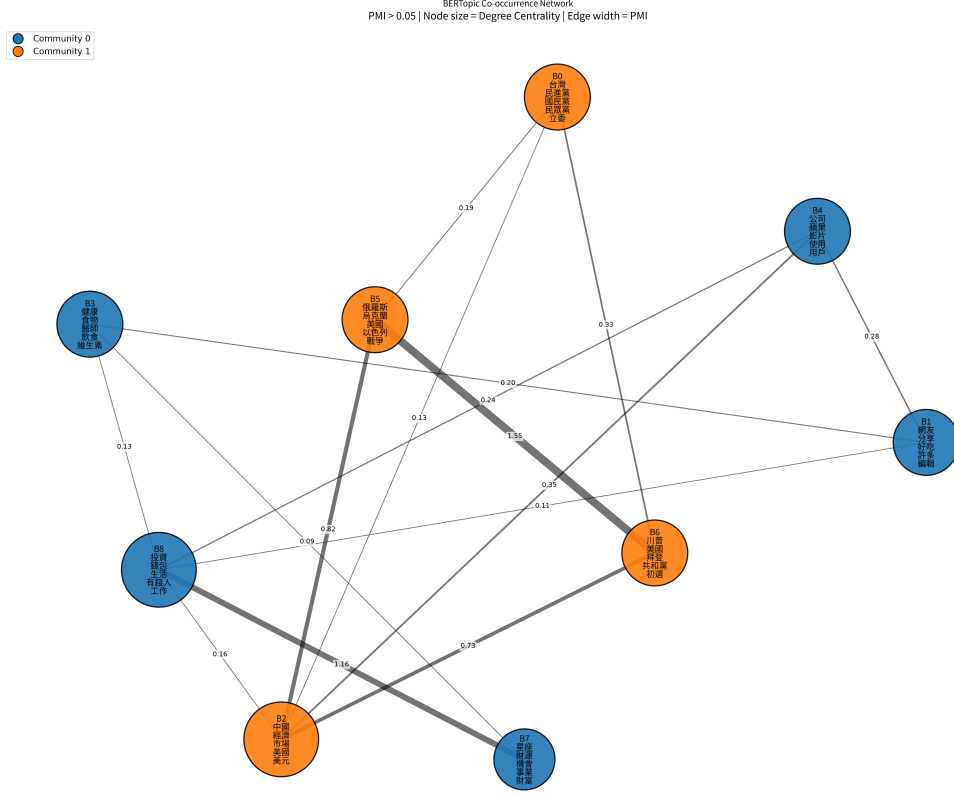


Figure 2: BERTopic topic co-occurrence network with $PMI > 0.05$.

4.5 Community Structure

The LDA network produces three communities:

- **LDA Community 0:** T5, T6, T7, T9, T10
- **LDA Community 1:** T3, T4, T8
- **LDA Community 2:** T1, T2

These communities can be interpreted as:

- Lifestyle, Economy, and International Affairs;
- Politics, Government, and Public Affairs;
- Lifestyle, Food, and Health.

The BERTopic network produces two communities:

- **BERTopic Community 0:** B1, B3, B4, B7, B8
- **BERTopic Community 1:** B0, B2, B5, B6

The first BERTopic community combines lifestyle, health, technology, and personal finance topics. The second community is dominated by politics, economics, and international affairs.

Table 4: Community-level interpretation of the two models.

Model	Community	Topics	Interpretation
LDA	C0	T5, T6, T7, T9, T10	Lifestyle, Economy, International Affairs
LDA	C1	T3, T4, T8	Politics, Government, Public Affairs
LDA	C2	T1, T2	Lifestyle, Food, Health
BERTopic	C0	B1, B3, B4, B7, B8	Lifestyle, Health, Technology, Personal Finance
BERTopic	C1	B0, B2, B5, B6	Politics, Economy, International Affairs

4.6 Cross-model Topic Assignment

The document-level comparison between LDA and BERTopic resulted in an Adjusted Rand Index (ARI) of 0.1597 and a Normalized Mutual Information (NMI) score of 0.3465.

The relatively low ARI indicates limited agreement between the exact cluster assignments produced by the two models. The NMI score indicates that the two solutions nevertheless share a measurable amount of information.

This difference is expected because the two approaches use substantially different representations. LDA is based on word distributions and assigns documents to latent probabilistic topics, whereas BERTopic relies on contextual document embeddings and clustering. Therefore, similar semantic themes do not necessarily result in identical document-level assignments.

5 Discussion

The results demonstrate several important differences between LDA and BERTopic.

First, LDA produces a relatively structured and interpretable topic distribution. The dominant-topic representation assigns every document to one topic, which makes topic counts and topic purity straightforward to interpret. Several LDA topics show high purity, particularly those related to food, health, and general lifestyle content.

BERTopic produces a different type of thematic structure. Its topics are often more semantically specific, particularly in areas such as international conflicts, technology, personal finance, and astrology. The ability to associate documents with multiple topics also produces a less exclusive representation of document themes.

Second, the network analysis reveals a denser structure for BERTopic. With the same PMI threshold, the BERTopic network contains more than twice as many edges as the LDA network. This suggests that BERTopic captures more connections between topics at the document level. However, the community structure is not simply a reflection of network density. LDA produces three communities, while BERTopic produces two broader communities.

The community-level results show an interesting common pattern. Both models separate a broad political/international domain from a lifestyle-oriented domain. In particular, the BERTopic community containing B0, B2, B5, and B6 brings together domestic politics, economics, and international affairs. Similarly, the LDA communities contain related political and international topics. This suggests that despite their different topic assignments, the two models identify several comparable high-level semantic dimensions.

Finally, the ARI and NMI results indicate that topic-level correspondence is limited but not absent. The relatively low ARI suggests that the two models should not be expected to produce interchangeable document clusters. Instead, their differences can be interpreted as complementary representations of the same corpus.

6 Final Model Comparison

The overall comparison is summarized in Table 5.

Table 5: Final comparison of LDA and BERTopic.

Metric	LDA	BERTopic
Topics	10	9
Documents	5764	5764
Average topics/document	1.00	1.96
PMI > 0.05 edges	7	15
Communities	3	2
Largest topic share	18.70%	27.43%
Average purity	69.03%	–
ARI		0.1597
NMI		0.3465

Overall, LDA provides a more discrete and probabilistically interpretable topic structure, while BERTopic provides a more flexible and semantically connected representation. The network analysis further demonstrates that these differences extend beyond individual topics to the organization of higher-level thematic communities.

7 Conclusion

This study compared LDA and BERTopic for topic discovery in a Chinese news corpus of 5,764 documents. Although both approaches identified broadly similar thematic dimensions, their topic structures differed considerably.

LDA produced 10 topics and a more exclusive document representation, with an average of one dominant topic per document and an average topic purity of 69.03%. BERTopic produced 9 topics and a more flexible representation in which documents were associated with an average of 1.96 topics. BERTopic also produced a denser PMI network, with 15 edges compared with 7 for LDA.

At the community level, both approaches identified a broad distinction between political/international content and lifestyle-oriented content, although they organized these themes differently. The relatively low ARI (0.1597) and moderate NMI (0.3465) indicate limited agreement at the document-assignment level.

Taken together, the findings suggest that LDA and BERTopic should be viewed as complementary rather than interchangeable methods. LDA provides a relatively transparent probabilistic representation of topic membership, while BERTopic captures broader semantic relationships through contextual embeddings. Combining topic modeling with PMI-based network and community analysis provides an additional perspective for examining how individual topics are connected and organized into higher-level thematic structures.